

ANDREW YING

PERSONAL INFORMATION

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EDUCATION

Johns Hopkins University

Baltimore, MD

M.S.E. Biomedical Engineering (GPA: 3.97)

August 2024 - May 2025

B.S. Biomedical Engineering & Computer Engineering (GPA: 3.97)

August 2021 - May 2025

SELECTED COURSEWORK

Computer Integrated Surgery I & II

Fall 2024 – Spring 2025

Robot Devices, Kinematics, Dynamics, and Control

Spring 2025

Computer Graphics

Spring 2025

Machine Learning: Deep Learning

Fall 2024

Computer Vision

Spring 2024

Signals & Systems

Spring 2024

Methods in Nucleic Acid Sequencing Lab

Spring 2024

Genomics Data Visualization

Spring 2024

Mathematical Statistics

Fall 2023

Biochemistry and Molecular Engineering

Fall 2022

HONORS AND AWARDS

John Boswell Whitehead Award

May 2025

- Presented annually for outstanding achievements in electrical and computer engineering by an undergraduate student, earning the highest GPA in the department

Dean's List

January 2022 – May 2025

- Earned by students who achieve a term grade point average of 3.5 or above

Hodson Trust Scholarship

August 2021 – May 2025

- Awarded on the basis of academic and personal achievement and leadership to fewer than twenty incoming first-year students each year through the Hodson Trust

RESEARCH EXPERIENCE

Laboratory for Computational Sensing + Robotics, Johns Hopkins University

Baltimore, MD

Graduate Researcher under Dr. Adnan Munawar

January 2025 – Present

- Developed virtual reality simulator using ROS, Chai3D, and PyBullet Physics for transsphenoidal surgery
- Implemented endoscopic camera with radial and tangential distortion using OpenGL and GLSL shaders
- Segmented drillable facial bone and flexible nasal passage in 3D Slicer for realistic soft body collisions

Kadoch Lab, Harvard Medical School/Dana-Farber Cancer Institute

Boston, MA

Research Intern

May 2024 – May 2025

Research Assistant

January 2026 - Present

- Analyzed RNA-seq, ATAC-seq, and CUT&RUN data to characterize the role of the BAF complex in CBFA2T3-GLIS2 fusion positive Acute Myeloid Leukemia
- Trained random forest models to identify biomarkers associated with cancer dependency on BAF complex
- Generated Kaplan-Meier curves to investigate clinical impact of ncBAF and SAGA complex interactions

Precision Care Medicine Project, Johns Hopkins University

Baltimore, MD

Team Member under Dr. Nitish Thakor

August 2024 – May 2025

- Analyzed stroke patient gait metrics produced by pressure-sensing insole to quantify therapy effectiveness
- Identified clinically relevant foot regions and assessed metric interpretability and ease-of-adoption based on literature review and clinician input
- Compared left-right foot asymmetries and inter-subject variability, and ranked each metric's predictive value

Biomedical Engineering Design Team, Johns Hopkins University

Baltimore, MD

Design Team Member under Dr. Hien Nguyen

March 2022 – May 2024

- Designed AI surgical training feedback and analytics platform using Flutter to improve resident education
- Interviewed clinicians, residents, and researchers to understand stakeholder needs and design criteria
- Conducted market potential research and competitive analysis and pursued Report of Invention with Johns Hopkins Tech Ventures
- Received \$1000 Catalyst Award from the Hopkins Office of Undergraduate Research

Popel Lab, Johns Hopkins University School of Medicine

Baltimore, MD

Undergraduate Researcher under Dr. Alberto Ippolito

October 2022 – May 2023

- Implemented neural network using PyTorch and NumPy to predict response to PD-L1 immunotherapy in triple negative breast cancer patients
- Expanded lab's quantitative systems pharmacology model to handle missing inputs and stochasticity using constant value, linear regression, and neural network-based imputation

Procter & Gamble Intelligent Automation Collaborative

Cincinnati, OH

Team Leader

June 2021 – August 2021

- Developed three attachments for the Swiffer to improve its collection of hair and cereal particles
- Managed order lists and requested manufacturing services from Kinetic Vision
- Presented attachments to Senior Director of Research & Development, Su Chang

Upper Airway Biomechanics Lab, University of Cincinnati College of Medicine

Cincinnati, OH

Student Researcher under Dr. Liran Oren

June 2020 – August 2020

- Analyzed particle-image velocimetry data from study on velopharyngeal insufficiency
- Produced and cleaned STL models of dog tracheas from CT scans using MeshMixer
- Created a PyQt5 GUI program using the Klatt Synthesizer for the lab's demo on vocal formants

TEACHING EXPERIENCE

Teaching Assistant

Biomedical Data Science

Linear Signals and Systems

Baltimore, MD

August 2024 – October 2024

October 2024 – December 2024

- Led section and taught 25 students exploratory data analysis in Python using Pandas and Numpy
- Answered questions regarding machine learning using scikit-learn and PyTorch during office hours
- Graded Jupyter Notebook assignments and provided detailed and constructive feedback

Lab Manager

Biomedical Engineering and Design

Structural Biology Lab

Baltimore, MD

August 2022 – December 2022

January 2023 – May 2023

- Gave assigned freshman group feedback on health inequity presentation and Arduino project lab report
- Taught freshman about using electrical circuits to model fluid flow during cardiovascular system module
- Guided freshman group through protein visualization and mutagenesis analysis in PyMOL

LEADERSHIP AND VOLUNTEERING EXPERIENCE**MedTech Network (MTN)**

Vice President: Workshops Committee

Baltimore, MD

September 2021 – May 2025

- Marketed and organized over 20 technical workshops on CAD, 3D printing, and Python for student body
- Taught 40 8th grade students about Arduino pulse sensor and breadboards at Science Olympiad Stem Expo

MedStar Union Memorial Hospital

Guest Ambassador Volunteer

Baltimore, MD

November 2022 – May 2023

- Provided comfort care and addressed patient needs in Cardiovascular ICU
- Performed rounds, made conversation with patients, and relayed their concerns to nurses
- Assisted nurses and physicians labeling TC-52 phones and refilling ultrasonic gel in patient rooms

Baltimore First

Volunteer

Baltimore, MD

September 2021 – December 2023

- Monitored chat and promoted atmosphere for Club 1111, a virtual nightclub for people with disabilities
- Cleared invasive ivy from urban green spaces and neighborhoods for Baltimore Green Space
- Organized used clothes and helped patrons select items at Paul's Place

PUBLICATIONS

H. Ishida*, **A. Ying***, A. S. Ding, Y. Heo, J. Wang, S. Vedula, W. Ishida, R. Taylor, M. Ishii, A. Munawar. "TRIDENT: Transsphenoidal Surgical Simulator for Realistic Interactive Drilling and Endoscope Navigation Training," 2026.

*These authors contributed equally

Manuscript submitted for publication at the *International Conference on Biomedical Robotics and Biomechatronics*

G. E. Boyle*, A. W. Ying*, S. Kaonis, L. Yao, J. L. Smith, C. J. Wong, L. Caven, N. Katiyar, M. Merrill, T. Hyelkma, Q. Le, T. Ishida, F. Sivandzade, M. Wither, **A. Ying**, K. E. Teichert, N. Garcia, S. Furuyama, R. Ries, I. Bernstein, S. Meshinchi, M. Meers, B. Hadland, R. Rau, S. Lin, C. Kadoch, J. F. Sarthy. “Targeting a mSWI/SNF-G1 axis as a novel therapeutic strategy for aggressive CBFA2T3-GLIS2 AML.”

*These authors contributed equally

Manuscript in preparation.

A. W. Ying, **A. Ying**, C. Kadoch. “Integrative small molecule and CRISPR screening reveals pervasive, specialized mSWI/SNF dependencies.”

Manuscript in preparation.

PRESENTATIONS

A. Ying, “Transsphenoidal Surgical Simulator,” Neocis, Miami, FL, 2025.

A. Ying, “Transsphenoidal Surgical Simulator,” Computer Integrated Surgery II Poster + Demo Session, Baltimore, MD, 2025.

O. Wei, Y. Zhao, Y. Chen, A. Cheng, C. Chen, **A. Ying**, “Gait Analysis using a High-Resolution Spatiotemporal Pressure Sensing Insole,” Johns Hopkins Engineering Design Day, Baltimore, MD, 2025.

O. Wei, Y. Zhao, Y. Chen, A. Cheng, C. Chen, **A. Ying**, “Gait Analysis using a High-Resolution Spatiotemporal Pressure Sensing Insole,” Institute for Computational Medicine Night, Baltimore, MD, 2025.

E. Rocher, M. Song, **A. Ying**, I. Sharma, K. Kampta, J. Kaur, C. Shin, P. Jhala, “SimMetric: Improve Every Day,” Whiting School of Engineering Advisory Board Meeting, Baltimore, MD, 2023.

E. Rocher, M. Song, **A. Ying**, I. Sharma, K. Kampta, J. Kaur, C. Shin, P. Jhala, “Improving Robotic Surgery Training,” Johns Hopkins Engineering Design Day, Baltimore, MD, 2023.

A. Ying, K. Betts, L. Early, D. Liu, J. Chan, “Intelligent Automation Collaboration: Gate Review,” Procter & Gamble, Cincinnati, OH, 2021.

TECHNICAL SKILLS

Software: C, C++, R, Python, Java, MATLAB, Flutter, Linux, Slurm, ROS, GLSL, PyTorch, Pandas, Matplotlib

Manufacturing: 3D Printing, Manual Mill, Laser Cutter, Plasma Router